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Studierichting: Informatica

Oefeningenreeks 6B nr. 17.5

$$\begin{aligned} s_n &= \sum_{k=1}^n (2k-1) 2k (2k+2) \\ &= \sum_{k=1}^n (2k-1) (4k^2 + 4k) \\ &= \sum_{k=1}^n (8k^3 + 4k^2 - 4k) \\ &= 8 \sum_{k=1}^n k^3 + 4 \sum_{k=1}^n k^2 - 4 \sum_{k=1}^n k \\ &= 8 \cdot \frac{n^2 (n+1)^2}{4} + 4 \cdot \frac{n (n+1) (2n+1)}{6} - 4 \cdot \frac{n (n+1)}{2} \\ &= 2n^2 (n+1)^2 + \frac{2n (n+1) (2n+1)}{3} - 2n (n+1) \\ &= 2n (n+1) \left(n(n+1) + \frac{2n+1}{3} - 1 \right) \\ &= 2n (n+1) \left(\frac{3n(n+1) + 2n+1-3}{3} \right) \\ &= \frac{2n (n+1)}{3} (3n^2 + 3n + 2n + 1 - 3) \\ &= \frac{2n (n+1)}{3} (3n^2 + 5n - 2) \\ &= \frac{2n (n+1)}{3} (3n-1) (n+2) \\ &= \frac{2n (n+1) (n+2) (3n-1)}{3} \end{aligned}$$

Omdat $n (n+1) (n+2)$ een product is van 3 opeenvolgende getallen, is één van die factoren deelbaar door 3.

$$\Rightarrow s_n \in \mathbb{N}$$

References